



Total Marks: 60 (20×3)

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet
- Use activation voltage $V_\gamma(\text{diode}) = 0.6 \text{ V}$, $V_\gamma(\text{transistor}) = 0.5 \text{ V}$, $V_{BE}(\text{forward active}) = 0.7 \text{ V}$ and $V_D(\text{conducting voltage of Diode}) = 0.7 \text{ V}$, for all the questions.
- There is no need to draw circuits in your answer sheet.

Name:	ID:	Section:
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Question 1

The **figure 1** on the left is a **square wave generator** constructed with a Schmitt trigger. The generator has an input and output waveform depicted by the **figure 2**. [20]

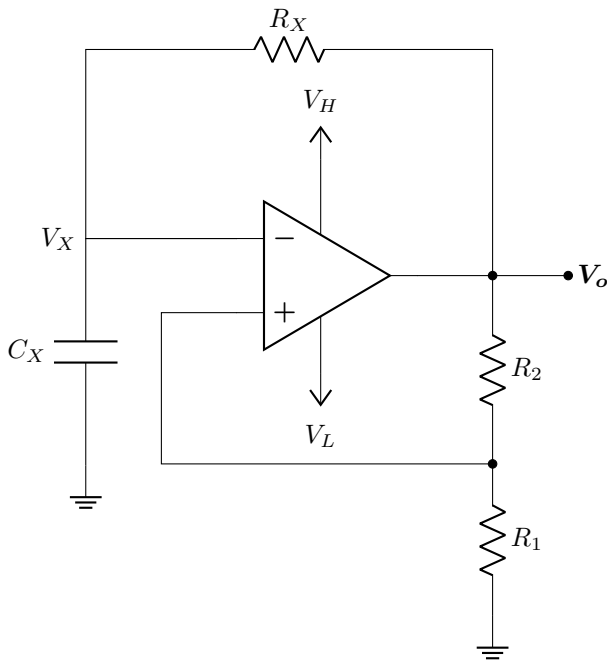


Figure 1

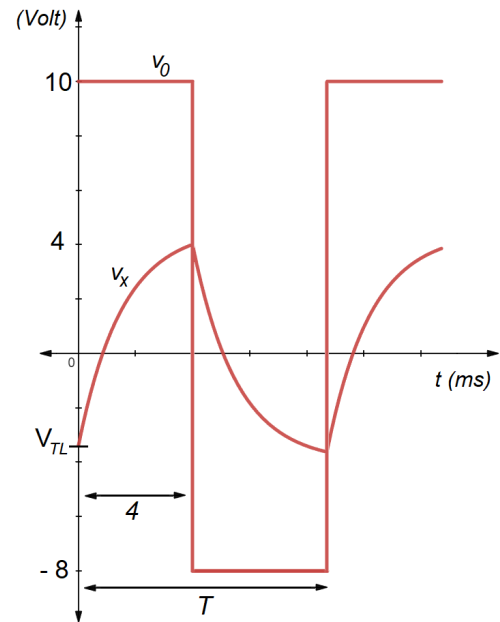


Figure 2

CO3	(a)	Identify whether the Schmitt Trigger in figure 1 is inverting or non-inverting .	[2]
CO3	(b)	Find the Hysteresis Width of the used Schmitt trigger.	[6]
CO3	(c)	Using $R_1 = 12 \text{ k}\Omega$, show that R_1 is smaller than R_2 .	[4]
CO3	(d)	Determine the duty cycle of the square wave generator circuit. (<i>hint: R_x and C_x values are not required for this part</i>)	[8]

Question 2

Despite the efforts of the last 2 months, Mr. Motlob is still stranded on the deserted island. He has now managed to find and set up an ECL circuit as shown in the **figure 3**. He knows that the **logical Low** for v_{O1} voltage is **2.5 V**. He has a bulb circuit that he attaches to v_{O2} .

The bulb circuit gives a

1. bright light when the $v_{O2} > 4V$
2. dim light when $2.8 \leq v_{O2} \leq 4V$.

CO1	(a)	Determine the value of i_E [Hint: consider v_{in} is logically Low and so Q_R is on, Q_1 is off and v_{O1} is low].	[8]
CO1	(b)	If v_{in} is logic High , determine the maximum value of R_{C1} , so that Mr. Motlob gets a dim light.	[8]
CO1	(c)	Find out the value of Reference voltage V_R [Hint: Reference voltage is the average of logic Low and logic High values]	[4]

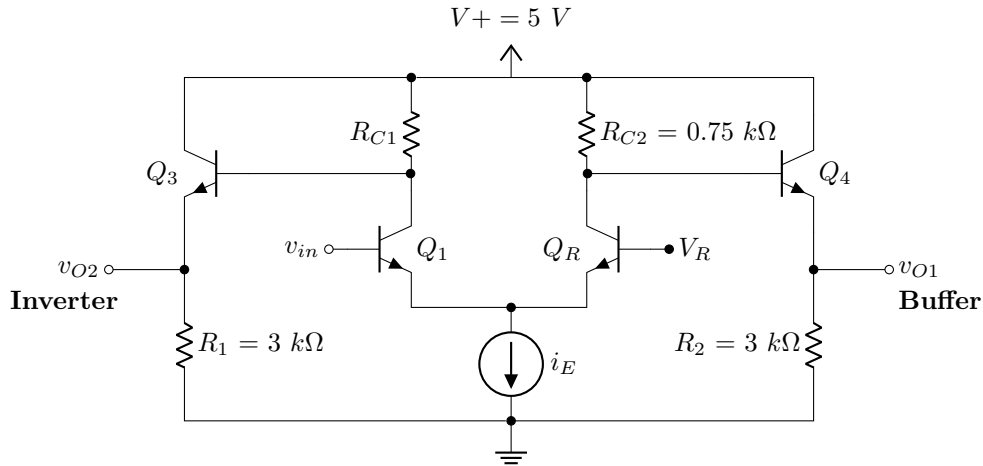


Figure 3

Question 3

A dual slope A/D converter shown in figure 4 has $V_{REF} = 5.12 \text{ V}$, and an N-bit counter that outputs N-bit representation of the input signal. A 2 MHz clock is used for the clock and counter circuits. [20]

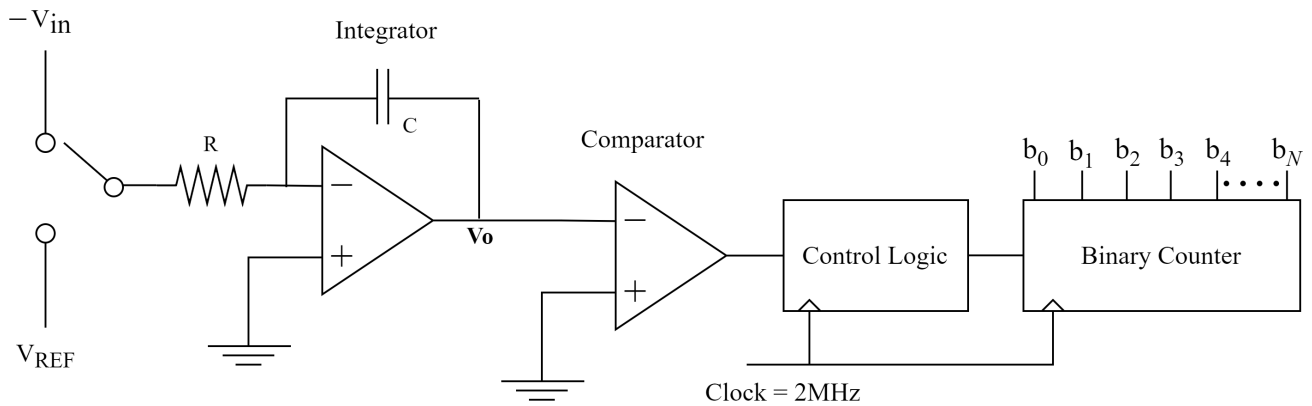


Figure 4

CO3	(a)	Determine the input voltage range. If step size is 0.02 V , determine the total number of steps and the number of bits used in this ADC.	[7]
CO3	(b)	For input voltage, $V_{in} = 3 \text{ V}$, determine the ADC output count. Calculate the time required to get this output.	[7]
CO3	(c)	Draw V_o vs time graph for the input voltage V_{in} mentioned in (b). If $V_{in} = 4 \text{ V}$, how the graph will change? Explain briefly.	[6]

Question 4

For the given TTL circuit in **figure 5**, common emitter current gain $\beta_F = 30$, $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.1V$, and reverse common emitter current gain of the transistors $\beta_R = 0.1$. Assume all inputs are **logical High** or **6V**. [20]

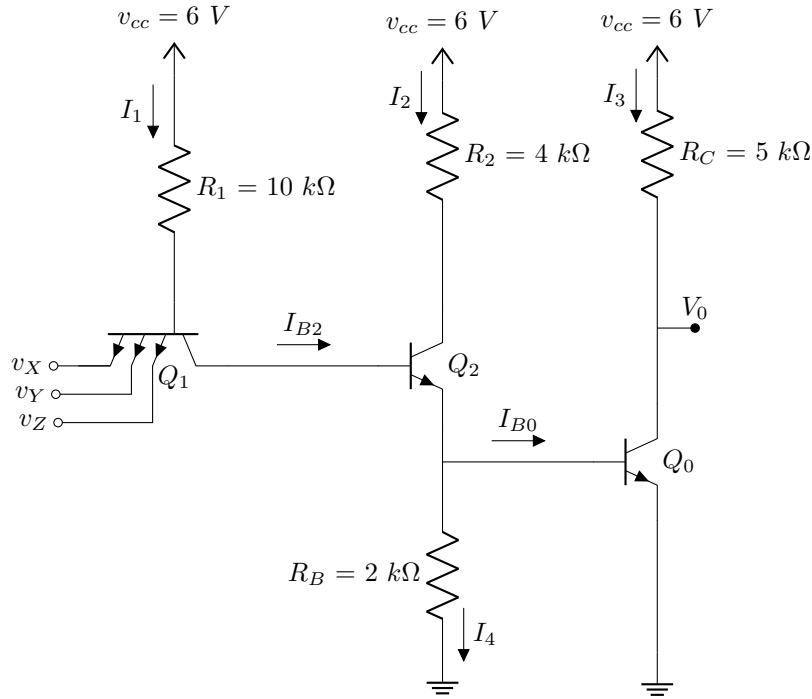


Figure 5

CO1	(a)	Calculate all the currents labeled in the above circuit.	[6]
CO1	(b)	If one of the emitters of the Q_1 transistor is removed, what will be the operating mode of Q_0 ? Justify your answer.	[6]
CO1	(c)	Design a CMOS logic circuit to implement the logic function, $F = XY + Z$. Assume that the signals X, Y, and Z are available. (<i>You must show both PMOS and NMOS networks</i>).	[8]